From single to multiple drivers – combined effect of temperature, pH and lithium on the larvae of the sea urchin *Paracentrotus lividus*

Abstract

1. Introduction

Anthropogenic activities lead to ocean changes in a host of environmental drivers threatening ecosystems worldwide (Pörtner et al., 2021). These threats include global changes such as ocean warming (OW) and acidification (OA), compounded by local changes such as pollution, habitat destruction, and resource over-exploitation (Boyd et al., 2018; IPCC, 2014). Understanding and resolving the cumulative impact of multiple drivers is one of the UN Decade's main challenges. Multiple drivers act simultaneously and often combine in non-linear ways, producing outcomes that are more severe and complex than from individual drivers alone. These aspects are neglected in the tools currently used for management and policy. The main reason is the lack of scientific understanding of the cumulative biological impact of multiple drivers due to conceptual and technical complexities. Therefore, these tools strongly underestimate the impacts**,** with the risk of measures that are not drastic enough to ensure a sustainable and healthy ocean.

Experimental studies on the biological impacts of the exposure to single drivers, and their combined effects has been the subject of many recent studies (for a review, see Boyd et al., 2018). At present, attempts to resolve the impact of global and local drivers are mostly performed using scenario-based cross-design experiments. These do not allow identifying non-linear effects or interactions and statistical interactions are often confused with driver interactions (Boyd et al., 2018). Therefore, the reported nature of interaction and combined effects varies between studies and is often based on misconceptions (see Crain et al., 2008; Stockbridge et al., 2020; Tekin et al., 2020 for contrasting reported interactions).

Resolving the biological effects of temperature (Byrne et al., 2022) and pH (Caiollon et al., 2023), alone or in combination, are highly relevant for global change studies. Their response curves over a large gradient are commonly unimodal. Different dose-response curves are traditionally observed for toxicants. For example, lithium from industry and pharmaceuticals are running off into the marine environment and is a concern especially in urbanized coastal areas (Ruocco et al., 2016). Response curves over toxicant gradients are often described as sigmoidal, with thresholds below which adverse effects are not expected to occur (Bellas, 2008; Morroni et al., 2023; Philibert et al., 2021).

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Because of practical limitations, multi-driver studies are often limited to two or three treatment levels per driver, using an ANOVA-based experimental design of 'control' and 'future' conditions. Not only does this approach often ignore the high level of present natural variability experienced by the tested species or ecosystems (e.g. Vargas et al., 2017, 2022), but neither does it allow for an evaluation of potential interactions. Many authors assume linearity of the response and confuse statistical interaction with more complex non-linear driver interactions (Burgess et al., 2021; Kreyling et al., 2018). There is an increased awareness of the drawbacks of this experimental design (e.g. Collins et al., 2022; Kreyling et al., 2014), leading to different experimental designs and strategies (Carmichael et al., 2025).

The aim of our study is to test a strategy that combines (i) the resolution of the shape of the response curves of 3 drivers (temperature, pH, and lithium); (ii) use these performance to project their combined impacts under a concentration-addition hypothesis; and (iii) evaluate the projections using a fully crossed design (2x2x2=8 treatments). This allows to identify potential interactions (non-additive effects) by comparing the outcome of the multiple driver experiment with the results of a non-linear additive model (e.g. Espinel-Velasco et al., 2025).

The implementation of such complex experiments is only realistic by using simple assays that are well-established and allow for the collection of robust data relatively fast. Sea urchins are such model organisms due to their ecological importance in benthic ecosystems, their sensitivity to environmental change, and their well-established use in ecotoxicology and climate change research (Agnetta et al., 2015; Bellas, 2008; Cuthbert et al., 2021; Dupont et al., 2010; Pagano et al., 2017). We assessed the effect of temperature, pH and lithium alone and combined on the growth rate and fitness of larvae of the purple sea urchin (*Paracentrotus lividus*) to answer two main questions:

(i) *What is the shape of their response curves for temperature, pH and lithium?*

(ii) *Are there interactions (non-additive effects) between the drivers?*

2. Materials and methods

2.1 Sampling and larval culture

Adults of the sea urchin *Paracentrotus lividus* were collected in Cap d'Ail, France, at 2-4m depth by snorkelling in October 2022. Organisms were kept in tanks with aerated seawater and transported within 30 minutes to the Marine Environment Laboratories of the International Atomic Energy Agency (IAEA) in Monaco. Animals were acclimated and maintained for 24h in an aquarium with flowing natural seawater (average $pH_T = 8.1$, temperature = 20°C, salinity = 38, dissolved oxygen? (x%)). Spawning was triggered by injection of 1mL 0.5M KCl in seawater across the peristomial membrane. Sperm was collected dry whereas eggs were collected into 50 mL beakers filled with filtered seawater. Gametes from one male and one female were fertilized (20 µL of activated sperm in 1L of egg solution). Three hours post-fertilization, two cell stages were transferred to 100 and 500 mL glass experimental bottles (~10 larvae mL^{-1}). A small subsample of eggs was fixed and the initial size of eggs (IS) was recorded (see below).

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2.2 Experimental design

Both single and multi-driver experiments were performed to evaluate the effect of pH, temperature and lithium alone and in combination. pH was manipulated by bubbling seawater with CO_2 till the target level was reached; lithium levels were reached using a 1000 ppm stock solution, and the temperature was controlled using water baths connected to a feedback system (IKS Aquastar, Germany). Experiments were carried out in closed 100 mL glass bottles, except for the temperature gradient experiment, which used 500 mL bottles. Embryos and larvae were exposed for 96 hours without addition of food. At the end of the exposure, seawater analyses and biological measurements were performed as described below.

Reference conditions (controls) of the experiments were chosen with the sampling site's average level in mind: 20°C, pH 8 and lithium 0.18 ppm, and the ranges of the treatment levels chosen to extend beyond the natural variability for October/November of 15°C – 23°C and pH 8.15 – 7.97 (Gattuso et al., 2021), as well as providing a lithium range from ocean average up to heavily contaminated seawater (Barbosa et al., 2023).

The single driver experiment (experiment #1) was carried out in 15 bottles, where there were five bottles (treatment levels) for each driver, providing a range of temperature (15°C to 30°C), pH (pH 8 to 7.45) and lithium (0.18 to 20 ppm) treatments in regression-based designs (Figure 1a). The multiple driver experiment (experiment #2) used two levels of each of these drivers in a full crossed design to form eight treatments, performed in duplicates (Figure 1a).

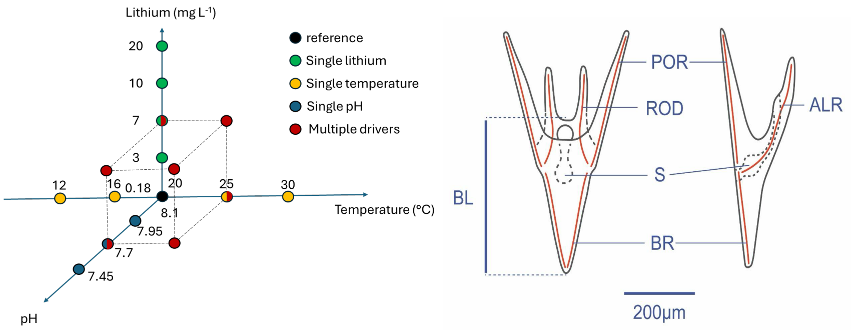

Figure 1. a) An overview of the single and multi-driver treatment levels, showing the target levels of the three drivers for all 23 treatments. b) *Paracentrotus lividus* pluteus larvae morphology and morphometric variables. BL: body length, BR: body rod, ALR: antero lateral rod, and POR: post oral rod.

2.3 Biological measurements

After 96 hours, a 10 mL sample was taken from each bottle and fixed with 4% formaldehyde. Larvae were counted for density and photographed under a 10x microscope (Brand of the microscope). Ten larvae were randomly selected from each treatment and body length (BL) measured using the Fiji software (Schindelin et al., 2012) (Figure 1b). The diameter of the eggs at the beginning of the experiment (Initial egg size, IS) was also measured. No change in density was detected except in the highest temperature treatment.

Larval growth rate (GR in µm*day^{-1}) was calculated for each aquarium as the difference between BL (µm) and IS (µm) divided by exposure time (3 days).

2.4 Data analyses

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Commented [SD4]: It seems that you do not calculate a mortality rate. If these data are not presented, no need to mention in the methods

Commented [EPJH4R2]: I made a basic mortality rate calculation (initial-final density)/days.
The highest temp treatment is the only one with clear mortality

Commented [EPJH4R3]: mention that no mortality was identified except in one treatment. Methods.

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The consistency in growth rate between treatments in common in the single- (experiment #1) and multiple-driver (experiment #2) experiments was evaluated with one-way ANOVAs and t-tests. A Shapiro-Wilks test indicated a modest deviation from normality for the residuals for some of the tests, but these tests are robust to such deviations (Glass et al., 1972; Knief & Forstmeier, 2021).

For each driver in the experiment #1, the shape of the response curve for growth rate was evaluated using linear models, second degree polynomial models, general additive models (GAMs), logit models and exponential models. There were no severe violations of assumptions for the models. The performance of the models was evaluated by comparing their R^2, standard error and AIC scores, and the appropriate model chosen for each driver based on these metrics. Other response curve models were also explored using the rTPC package in R (Padfield et al., 2021), but none fit well.

For the experiment #2, the effect of each driver on growth rate was compared when considered as a single driver (e.g. only high temperature) or when combined with one or two of the other drivers (e.g. any combination with low pH and/or high lithium). Wilcoxon rank-sum tests were used to test for statistical significance in the comparison of each pair.

For a comparison between the 3 different drivers, a driver index was calculated for each driver (temperature, pH and lithium) as the weighted (α_{driver}) deviation from the reference condition (Δ_{driver}):

$$DI_{driver} = \alpha_{driver} * \Delta driver$$

α was estimated for each driver by aligning the three single driver performance curves along the driver index axis. These curves were subsequently merged into one performance curve modelled from all the single driver data.

Assuming a similar mode of action between drivers and using a concentration-addition approach, the DI was then calculated for the multiple driver experiments as: $DI = \alpha_{Temp} DI_{Temp} + \alpha_{pH} DI_{pH} + \alpha_{Li} DI_{Li}$

The data from the single- and multiple driver experiments were compared on this driver index. Any deviation from the model built from the single driver experiment reveals an interaction between the drivers.

2.5 Seawater parameters

Temperature data were collected from continuous sensors (brand of probes). pH_T (pH on the total scale) of the seawater was measured in each aquarium at the end of the experiment with a glass pH electrode (brand?) calibrated with TRIS buffer solution (Andrew Dickson, Scripps Institution of Oceanography, San Diego, USA). Total Alkalinity (TA) was measured using the automatic open cell potentiometric titration method (brand) following the protocol described by Dickson et al. (2007). pCO_2, calcite and aragonite saturation values were calculated from TA, pH_T, and temperature at a salinity of 38 psu using the R package *seacarb* (Gattuso et al., 2016), using the dissociation constants from Lueker et al. (2005).

Lithium samples were filtered through 1.0 µm polycarbonate filters and lithium concentration analyses in seawater samples were carried out by Inductively coupled plasma - optical emission spectrometry technique (ICP-OES; Plasma Quant 9100, Analytik Jena, Germany). Samples were acidified with ultrapure HNO_3 and diluted 10

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I spent a lot of time thinking about it and this tests a hypothesis that was never presented before.

It assumes that for each driver (e.g. pH), if you add others (e.g. temperature, lithium, or both) that does not change anything.

It is not a hypothesis that you can support with what we know except maybe for temperature if you assume that it is a strongly dominating driver and that the treatment selected for the two others are too weak to be detectable or matters.

This would be defendable but would need to be explained in the Introduction (and justified that effect size for temperature tend to be much more important than pH?) We may have request of better justification of the tested scenarios by the reviewers.

I would then only test that one:
Compare temperature as a single stressor with temperature with the others.

We can discuss this but these analyses should be justified by a hypothesis.

We can work on the rephrasing after agreeing on this

Commented [EPJH6R2]: Our landing was:
Let's include this analysis to show that there is an increased effect of multiple drivers, except at 24°C, indicating antagonistic effects there.

We then delve into those effects with the new SI graph

Commented [EPJH6R3]: I'd appreciate help with the wording of this. Because we want to keep it to say that there seems to be an antagonistic effect at 24°C, and that there is a larger effect of MS than SS in the other comparisons. But we are not doing it to test any null hypothesis that makes sense. So how do we justify doing this boxplot comparison in the first place?

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Commented [SD7]: Questions for Marc / Frankie

Commented [SD8]: Questions for Marc / Frankie times with ultrapure H_2O before analysis. The instrumental conditions are summarized in supplementary material (Table S1). The radial analysis mode was used for lithium enriched samples while the axial mode (more sensitive) was used for non-enriched seawater samples. The quantification was carried out by means of external calibration, using proper dilution of lithium stock standard (TraceCert®, 997 ± 4 mg/L, Sigma Aldrich, Switzerland) in 0.35 % ultrapure NaCl (ESI Elemental Scientific, USA), to assure matrix match with seawater samples diluted 10 times. In the absence of a seawater Reference Material certified for lithium (not available in the market), several spiked samples were analyzed to account for possible matrix effect and to evaluate the accuracy of results. The recoveries were always within acceptable range (90-110 %). The results are reported in mg/L (≈ ppm) together with respective expanded uncertainty (Ue, k=2).

3. Results

While exposed to the same treatments in the two experiments, larvae had no significant growth rate differences between the two experiments for pH and temperature (Figure 2). The lithium 7 ppm treatment had a higher growth rate in experiment #1 (74.5 um/day, SD=2.8) than in the experiment #2 (68.6 um/day, SD=6.6, Welch's t-test: t(24.6)=3.4, p= 0.0026). There was also a slight but significant difference between the reference treatments, again with a higher growth rate in experiment #1 (74 um/day, SD=4.6) than in experiment #2 (69.6 um/day, SD=5.9, Wilcoxon rank test r=0.32, p=0.0091), with one of the replicates in experiment #2 having lower growth rate than the other one (63.8 vs. 74.9 um/day).

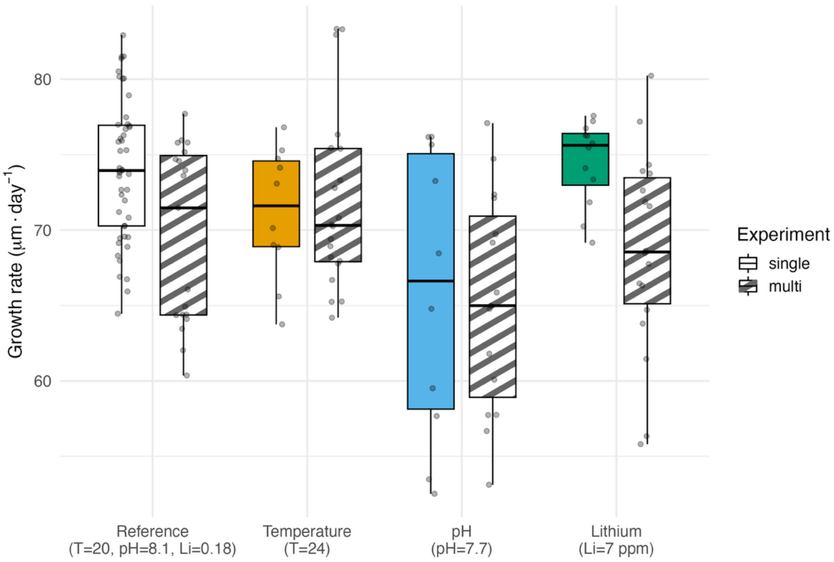
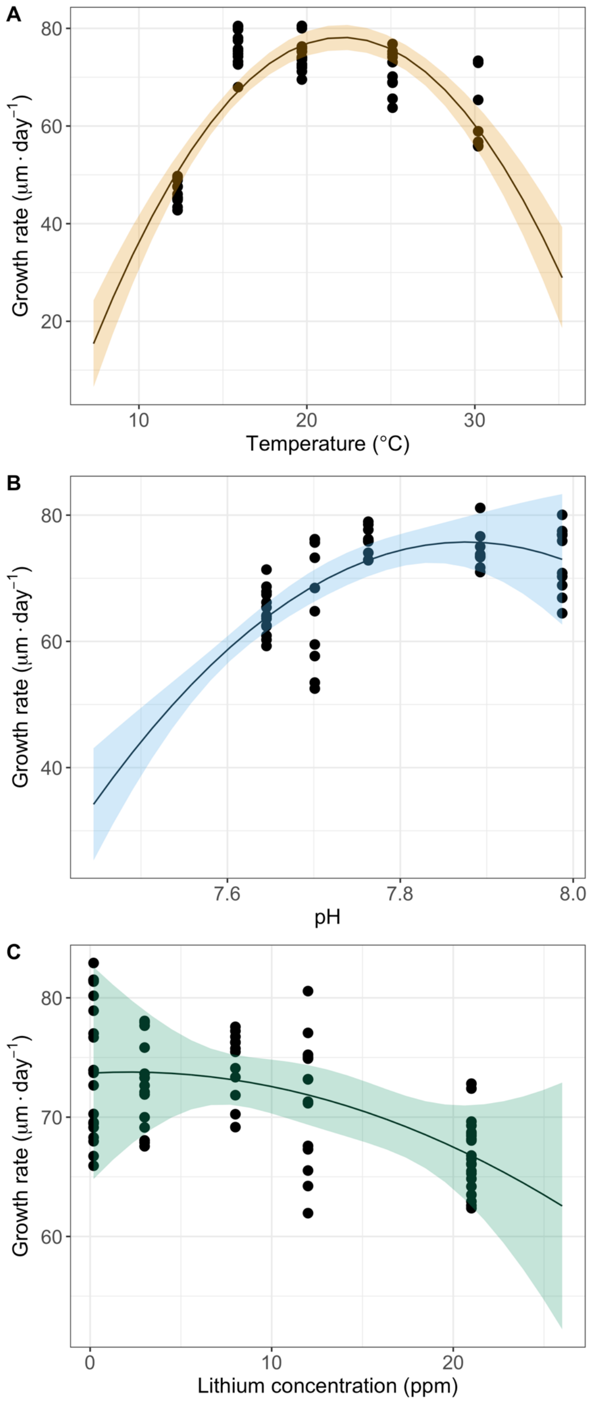


Figure 2. Growth rate comparison of the treatments in common to the two experiments. The reference conditions and lithium 7 ppm are the only significantly different comparisons.

The second-degree polynomial model was significant and the best fit for each drivers' growth rate data (Figures 3a-c, Table S2). For temperature, the curve was bell-shaped with a good fit (adj. $R^2 = 0.70$, $F_{(2, 54)} = 66.82$, $p<0.05$) and a tight confidence interval. Second degree polynomial model and GAM had comparable fits for the growth rate data from the pH treatments, but the polynomial was chosen with parsimony in mind. Compared to temperature, the polynomial model fit for pH had a lower fit and wider confidence interval (adj. R^2: 0.36, $F_{(2, 49)}=15.68$ $p=5.837e-06$). For lithium, the models had the poorest fit with the polynomial model being the best (adj. R^2: 0.30, $F_{(2,78)}=17.99$, $p=3.771e-07$), although the second-degree component was not significant. For all drivers, the reference treatments (20ºC, pH_T 8.0, lithium 0.18 ppm) had the highest GR.


A
B
C
Growth rate (µm · day^{-1})
Temperature (°C)
pH
Lithium concentration (ppm)
20
40
60
80
10
20
30
7.6
7.8
8.0
70
0

204

Figure 3a-c. Growth rate of *P. lividus* larvae across different (A) temperature, (B) pH_T, and (C) lithium concentrations. The solid lines indicate second degree polynomial models fitted to the datapoints and the ribbons the 95% confidence interval for each model.

The comparison of growth rate of larvae exposed to single or multiple drivers in the experiment #2 revealed (i) no significant differences at high temperature or high lithium when the larvae were exposed to those drivers alone or when combined with other drivers (Figures 2a and 2c), and (ii) a higher growth at low pH when combined with other drivers than when (Figure 2b) (see Table S3 for Wilcoxon tests).

Several α values were compared to synchronize and align the three single driver performance curves along the driver index axis. The best fits were 1, 28 and 0. 35 for temperature, pH and lithium, respectively (Figure 3a). These can be interpreted as a decrease by 1 pH unit relative to the reference pH 8.0 is corresponding to an equivalent effect on the temperature performance curve of a 28ºC increase relative to the reference 20ºC. A performance curve combining the 3 single drivers growth rate data was fitted after transformation into a driver index (Figure 3a), and the best fit was a second-degree polynomial curve (adj. $R^2 = 0.58$, $F(2, 187) = 129.43$, $p < 0.05$).

The 8 treatments of the multiple driver experiment were converted into a driver index following the concentration addition model and the growth rate data were plotted and compared with the polynomial model derived from the single driver experiment (Figure 3b). All the multiple driver treatments involving low pH were deviating from the polynomial model with higher growth rate than predicted and suggesting an antagonistic interaction.

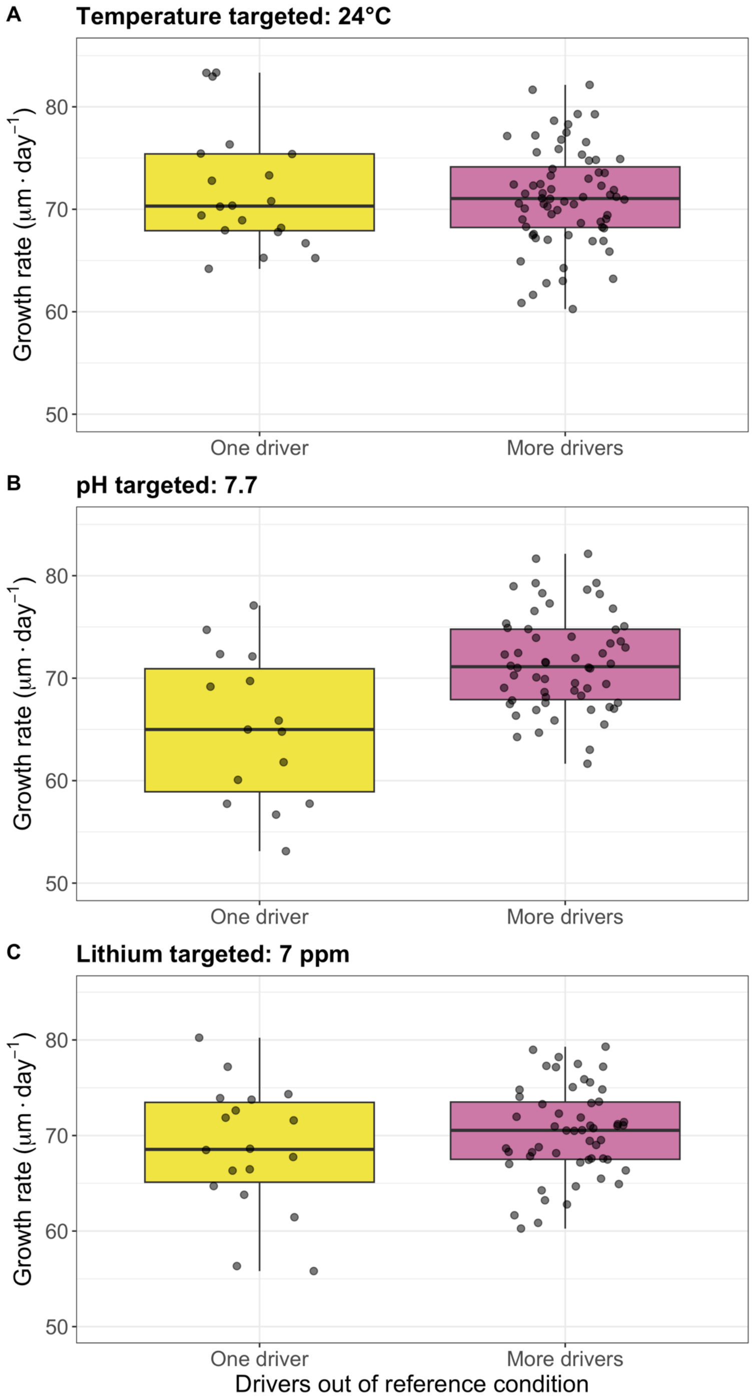
231 
A
Temperature targeted: 24°C
Growth rate (µm · day⁻¹)
80
70
60
50
One driver
More drivers
B
pH targeted: 7.7
Growth rate (µm · day⁻¹)
80
70
60
50
One driver
More drivers
C
Lithium targeted: 7 ppm
Growth rate (µm · day⁻¹)
80
70
60
50
One driver
More drivers
Drivers out of reference condition

Figure 2a-c. Comparison the growth rate of those of multi-driver treatments that had one driver out of reference conditions and those that additionally had more drivers out of reference conditions.

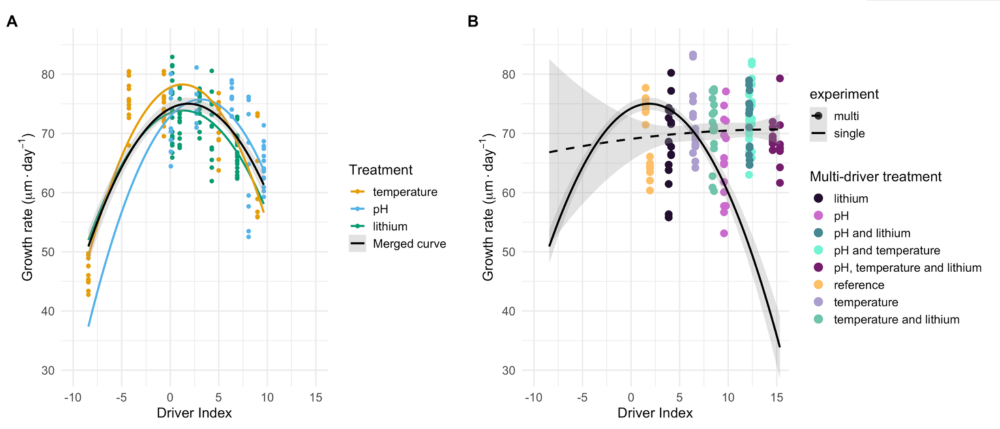


Figure 3a-b. A) The three single driver growth rate performance curves aligned on a common driver index axis, and a curve fit to all the single driver data combined. B) Comparison of the single driver curve with the multi-driver curve and points.

The parameters of the seawater physico-chemical parameters in the two experiments are summarized in the Table 2.

Table 2. Measured and calculated seawater parameters in each experimental treatment. pCO_2, DIC, Ω_{ca} and Ω_{ar} were calculated from measured TA, pH_T, temperature and a salinity of 38 psu. S=Single, M=Multiple, T=Temperature and Li=Lithium. Multi-driver duplicates have been merged.

Treatment	T (°C)	pH_T-*in situ*	Li (mg L^{-1})	TA (µmol kg^{-1})	pCO_2 (µatm)	DIC (µmol kg^{-1})	Ω_{ar}	Ω_{ca}
Single drivers								
12 °C	12.3	8.03	0.18	2548	461	2320	2.52	3.92
16 °C	15.99	8.01	0.17	2552	489	2299	2.81	4.33
20 °C	19.72	8.01	0.2	2557	479	2279	3.09	4.75
25 °C	25.08	8.00	0.19	2532	482	2214	3.57	5.40
30 °C	30.2	8.04	0.19	2575	435	2202	4.22	6.33
pH 8.1	19.72	7.99	0.19	2525	507	2259	2.98	4.57
pH 7.95	19.72	7.89	0.18	2519	654	2307	2.45	3.75
pH 7.7	19.72	7.76	0.18	2508	914	2358	1.88	2.88
pH 7.45	19.72	7.70	0.18	2502	1068	2379	1.65	2.53
pH 7.2	19.72	7.65	0.17	2511	1235	2410	1.48	2.26
0.18 ppm lithium	19.72	7.98	0.19	2516	513	2258	2.89	4.43
3 ppm lithium	19.72	7.99	3	2507	500	2245	2.92	4.48
7 ppm lithium	19.72	7.98	8	2507	515	2250	2.88	4.41
10 ppm lithium	19.72	7.98	12	2498	506	2239	2.88	4.42
20 ppm lithium	19.72	8.00	21	2464	474	2196	2.95	4.53
Multiple drivers								
MT20pH8.1Li0.18	20.8	7.97	0.18	2512	533	2256	2.88	4.40
MT20pH7.7Li0.18	20.6	7.67	0.19	2497	1149	2383	1.58	2.42
MT20pH8.1Li7	20.6	7.97	7.5	2482	523	2229	2.83	4.33
MT20pH7.7Li7	20.7	7.69	7.5	2471	1091	2350	1.62	2.48
MT24pH8.1Li0.18	25.5	7.97	0.18	2496	528	2201	3.34	5.04
MT24pH7.7Li0.18	25.5	7.76	0.18	2505	935	2323	2.24	3.39
MT24pH8.1Li7	25.4	7.97	7	2493	516	2194	3.37	5.09
MT24pH7.7Li7	25.6	7.76	7.5	2500	934	2318	2.25	3.39

4. Discussion

Research aiming at resolving the biological impacts of multiple drivers has increased rapidly over the last two decades. Within the field of ocean acidification alone, several thousands of articles are documenting experiments combining various driver combinations, often using relatively simple full-factorial designs. Despite this effort, there are few generalizable conclusions and contrasting results are quite common (e.g. Alter et al., 2024; Stockbridge et al., 2020). This can be attributed to inherent complexities in studying the combined effects of multiple drivers as these may have different modes of action and non-linear performance curves leading to difficulties in interpreting the results from full-factorial experiments (Boyd et al., 2018; Collins et al., 2022). Another reason is that simple full-factorial experiments are not adapted to resolve potential interactions between drivers as statistical interactions do not necessarily indicate driver interactions, defined as deviations from non-linear additivity.

In an ideal world, one could perform full-factorial experiments with a large number of treatments for each individual driver allowing to model the full response landscape and test different hypotheses to resolve interactions (e.g. a comparison of a concentration addition model with the landscape). These are often logistically difficult to implement, and alternatives should then be considered. We proposed and tested here such an alternative divided into 3 steps.

A first step towards identifying potential interactions between drivers is to understand the performance curve of each driver. In our experiment, the curves from the three drivers followed a similar unimodal second-degree polynomial model. This is the pre-requisite for the concentration addition hypothesis as the drivers are then expected to have the same mode of actions and similar biological responses (REF). Although all performance curves were second-degree polynomial in shape, the curve of temperature was the one best captured and was clearly bell shaped, as expected from literature (Byrne et al., 2022; Silva Romero et al., 2021). A similar unimodal shape has been found for phytoplankton responses to pH/pCO_2 (Collins et al., 2022; Paul & Bach, 2020). A sigmoidal curve is often expected for biological response to toxicants effects (Bellas, 2008; Morroni et al., 2023; Philibert et al., 2021). Our data did not allow to fully resolve the performance curves of pH and lithium. This can be performed in future work using treatments below and above the scenarios considered in our study and that the species will unlikely face, even in the worst-case scenarios of global change (Collins et al., 2022; Thibon et al., 2023).

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A second step is the calculation of an index combining the relative contribution of the individual driver following a methodology recently used by Espinel-Velasco et al. (2025). The calculation of this index was following a concentration addition model (Drescher & Boedeker, 1995; Liess et al., 2016; N. Wang et al., 2015). The index was calculated as the relative contribution of each driver relative to a change in temperature (in Celsius deviations from reference scale). The contribution of pH and lithium corresponded to an equivalent of 28°C for each pH unit change and 0.35°C for each lithium ppm change. This allowed to navigate along a shared unimodal response curve for the three drivers with a tipping point around 25°C, a 5°C deviation from the reference condition of 20°C. The third and last step of our strategy was to compare the outcome of the model based on single drivers with data from the full-factorial experiment with different combinations of the 3 drivers. Under the additivity hypothesis, these data should fit under the unimodal performance curve developed in step 2. We observed a deviation from that model for all the combinations involving pH, with the model over-estimating the impacts of those treatments. The treatments where only pH was out of reference had a higher growth rate than those that had more drivers simultaneously out of reference conditions (Figure 2b), revealing a true antagonistic interaction. Such interaction is somewhat surprising as previous studies showed that the combination of toxicants and low pH led to an increased stress (Wang et al., 2025). Understanding the mechanisms behind such interactions would require further work and a better understanding of each driver's mode of action and underlying physiological mechanisms affected (Boyd et al., 2018).

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Conclusion

Our strategy combining the resolution of individual performance curves and the calculation of an index under an additivity hypothesis is an excellent tool to identify true interactions between drivers. Our results allow to identify an antagonistic interaction between pH and the two other drivers (temperature and lithium). The next step consists in resolving the mechanisms of this interaction and incorporate it into a predictive model. This will allow to project the biological response to anthropogenically induced global and local changes under different scenarios and provide key information for management and local adaptation.

References

Agnetta, D., Badalamenti, F., Ceccherelli, G., Di Trapani, F., Bonaviri, C., & Gianguzza, P. (2015). Role of two co-occurring Mediterranean sea urchins in the formation of barren from Cystoseira canopy. *Estuarine, Coastal and Shelf Science*, *152*, 73–77. https://doi.org/10.1016/j.ecss.2014.11.023

Alter, K., Jacquemont, J., Claudet, J., Lattuca, M. E., Barrantes, M. E., Marras, S., Manríquez, P. H., González, C. P., Fernández, D. A., Peck, M. A., Cattano, C., Milazzo, M., Mark, F. C., & Domenici, P. (2024). Hidden impacts of ocean warming and acidification on biological responses of marine animals revealed through meta-analysis. *Nature Communications*, *15*(1), 2885. https://doi.org/10.1038/s41467-024-47064-3

Barbosa, H., Soares, A. M. V. M., Pereira, E., & Freitas, R. (2023). Lithium: A review on concentrations and impacts in marine and coastal systems. *Science of The Total Environment*, *857*, 159374. https://doi.org/10.1016/j.scitotenv.2022.159374

Bellas, J. (2008). Prediction and assessment of mixture toxicity of compounds in antifouling paints using the sea-urchin embryo-larval bioassay. *Aquatic Toxicology*, *88*(4), 308–315. https://doi.org/10.1016/j.aquatox.2008.05.011

Boyd, P. W., Collins, S., Dupont, S., Fabricius, K., Gattuso, J. P., Havenhand, J., Hutchins, D. A., Riebesell, U., Rintoul, M. S., Vichi, M., Biswas, H., Ciotti, A., Gao, K., Gehlen, M., Hurd, C. L., Kurihara, H., McGraw, C. M., Navarro, J. M., Nilsson, G. E., … Pörtner, H. O. (2018). Experimental strategies to assess the biological ramifications of multiple drivers of global ocean change—A review. *Global Change Biology*, *24*(6), 2239–2261. https://doi.org/10.1111/gcb.14102

Burgess, B. J., Purves, D., Mace, G., & Murrell, D. J. (2021). Classifying ecosystem stressor interactions: Theory highlights the data limitations of the additive null model and the difficulty in revealing ecological surprises. *Global Change Biology*, *27*(13), 3052–3065. https://doi.org/10.1111/gcb.15630

Byrne, M., Gall, M. L., Campbell, H., Lamare, M. D., & Holmes, S. P. (2022). Staying in place and moving in space: Contrasting larval thermal sensitivity explains distributional changes of sympatric sea urchin species to habitat warming. *Global Change Biology*, *28*(9), 3040–3053. https://doi.org/10.1111/gcb.16116

Carmichael, H., Warfield, R., & Yvon-Durocher, G. (2025). Reconciling Variability in Multiple Stressor Effects Using Environmental Performance Curves. *Ecology Letters*, *28*(1). https://doi.org/10.1111/ele.70065

Collins, S., Whittaker, H., & Thomas, M. K. (2022). The need for unrealistic experiments in global change biology. *Current Opinion in Microbiology*, *68*, 102151. https://doi.org/10.1016/j.mib.2022.102151

Crain, C. M., Kroeker, K., & Halpern, B. S. (2008). Interactive and cumulative effects of multiple human stressors in marine systems. *Ecology Letters*, *11*(12), 1304–1315. https://doi.org/10.1111/j.1461-0248.2008.01253.x

Cuthbert, R. N., Sidow, A., Frost, K. F., Kotronaki, S. G., & Briski, E. (2021). Emergent effects of temperature and salinity on mortality of a key herbivore. *Journal of Sea Research*, *177*, 102126. https://doi.org/10.1016/j.seares.2021.102126

Drescher, K., & Boedeker, W. (1995). Assessment of the Combined Effects of Substances: The Relationship between Concentration Addition and Independent Action. *Biometrics*, *51*(2), 716. https://doi.org/10.2307/2532957

Dupont, S., Ortega-Martínez, O., & Thorndyke, M. (2010). Impact of near-future ocean acidification on echinoderms. *Ecotoxicology*, *19*(3), 449–462. https://doi.org/10.1007/s10646-010-0463-6

Espinel-Velasco, N., Kvernvik, A. C., Hop, H., & Dupont, S. (2025). Combined effects of ocean acidification, warming, and salinity on the fertilization success in an Arctic population of sea urchins. *Scientific Reports*, *15*(1), 44090. https://doi.org/10.1038/s41598-025-27725-z

Gattuso, J. P., Alliouane, S., & Mousseau, L. (2021). Seawater carbonate chemistry in the Bay of Villefranche, Point B (France), January 2007-June 2023. (n.d.).

Gattuso, J.-P., Epitalon, J.-M., & Lavigne, H. (2016). *seacarb: Seawater carbonate chemistry with R. R package version 3.0.14* [Computer software].

Glass, G. V., Peckham, P. D., & Sanders, J. R. (1972). Consequences of Failure to Meet Assumptions Underlying the Fixed Effects Analyses of Variance and Covariance. *Review* *of Educational Research*, *42*(3), 237–288. https://doi.org/10.3102/00346543042003237

IPCC. (2014). *Climate Change 2014: Impacts, Adaptation and Vulnerability*.

Knief, U., & Forstmeier, W. (2021). Violating the normality assumption may be the lesser of two evils. *Behavior Research Methods*, *53*(6), 2576–2590. https://doi.org/10.3758/s13428-021-01587-5

Kreyling, J., Jentsch, A., & Beier, C. (2014). Beyond realism in climate change experiments: Gradient approaches identify thresholds and tipping points. *Ecology Letters*, *17*(1), 125. https://doi.org/10.1111/ele.12193

Kreyling, J., Schweiger, A. H., Bahn, M., Ineson, P., Migliavacca, M., Morel-Journel, T., Christiansen, J. R., Schtickzelle, N., & Larsen, K. S. (2018). To replicate, or not to replicate – that is the question: How to tackle nonlinear responses in ecological experiments. *Ecology Letters*, *21*(11), 1629–1638. https://doi.org/10.1111/ele.13134

Liess, M., Foit, K., Knillmann, S., Schäfer, R. B., & Liess, H.-D. (2016). Predicting the synergy of multiple stress effects. *Scientific Reports*, *6*(1), 32965. https://doi.org/10.1038/srep32965

Lueker, T. J., Dickson, A. G., & Keeling, C. D. (5 C.E.). Ocean pCO2 calculated from dissolved inorganic carbon, alkalinity, and equations for K1 and K2: Validation based on laboratory measurements of CO2 in gas and seawater at equilibrium. *Marine Chemistry*, *70*(1–3), 105–119. https://doi.org/10.1016/S0304-4203(00)00022-0

Morroni, L., Rakaj, A., Grosso, L., Flori, G., Fianchini, A., Pellegrini, D., & Regoli, F. (2023). Echinoderm larvae as bioindicators for the assessment of marine pollution: Sea urchin and sea cucumber responsiveness and future perspectives. *Environmental Pollution*, *335*, 122285. https://doi.org/10.1016/j.envpol.2023.122285

Padfield, D., O'Sullivan, H., & Pawar, S. (2021). *rTPC* and *nls.multstart*: A new pipeline to fit thermal performance curves in R. *Methods in Ecology and Evolution*, *12*(6), 1138–1143. https://doi.org/10.1111/2041-210X.13585

Pagano, G., Guida, M., Trifuoggi, M., Thomas, P., Palumbo, A., Romano, G., & Oral, R. (2017). Sea Urchin Bioassays in Toxicity Testing: I. Inorganics, Organics, Complex Mixtures and Natural Products. *Expert Opinion on Environmental Biology*, *06*(01). https://doi.org/10.4172/2325-9655.1000142

Paul, A. J., & Bach, L. T. (2020). Universal response pattern of phytoplankton growth rates to increasing CO_2. *New Phytologist*, *228*(6), 1710–1716. https://doi.org/10.1111/nph.16806

Philibert, D., Parkerton, T., Marteinson, S., & De Jourdan, B. (2021). Assessing the Toxicity of Individual Aromatic Compounds and Mixtures to American Lobster (*Homarus americanus*) Larvae Using a Passive Dosing System. *Environmental Toxicology and Chemistry*, *40*(5), 1379–1388. https://doi.org/10.1002/etc.4988

Pörtner, H.-O., Scholes, R. J., Agard, J., Archer, E., Arneth, A., Bai, X., Barnes, D., Burrows, M., Chan, L., Cheung, W. L. (William), Diamond, S., Donatti, C., Duarte, C., Eisenhauer, N., Foden, W., Gasalla, M. A., Handa, C., Hickler, T., Hoegh-Guldberg, O., … Ngo, H. (2021). *Scientific outcome of the IPBES-IPCC co-sponsored workshop on biodiversity and climate change* (Version 5). Zenodo. https://doi.org/10.5281/ZENODO.4659158

Ruocco, N., Costantini, M., & Santella, L. (2016). New insights into negative effects of lithium on sea urchin Paracentrotus lividus embryos. *Scientific Reports*, *6*(1), 32157. https://doi.org/10.1038/srep32157

Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., Tinevez, J. Y., White, D. J., Hartenstein, V., Eliceiri, K., Tomancak, P., & Cardona, A. (2012). Fiji: An open-source platform for biological-image analysis. *Nature Methods*. https://doi.org/10.1038/nmeth.2019

Silva Romero, I., Bruno, J. F., Silbiger, N. J., & Brandt, M. (2021). Local conditions influence thermal sensitivity of pencil urchin populations (Eucidaris galapagensis) in the Galápagos Archipelago. *Marine Biology*, *168*(3), 34. https://doi.org/10.1007/s00227-021-03836-9

Stockbridge, J., Jones, A. R., & Gillanders, B. M. (2020). A meta-analysis of multiple stressors on seagrasses in the context of marine spatial cumulative impacts assessment. *Scientific Reports*, *10*(1). https://doi.org/10.1038/s41598-020-68801-w

Tekin, E., Diamant, E. S., Cruz-Loya, M., Enriquez, V., Singh, N., Savage, V. M., & Yeh, P. J. (2020). Using a newly introduced framework to measure ecological stressor interactions. *Ecology Letters*, *23*(9), 1391–1403. https://doi.org/10.1111/ele.13533

Thibon, F., Weppe, L., Churlaud, C., Lacoue-Labarthe, T., Gasparini, S., Cherel, Y., Bustamante, P., & Vigier, N. (2023). Lithium isotopes in marine food webs: Effect of ecological and environmental parameters. *Frontiers in Environmental Chemistry*, *3*, 1060651. https://doi.org/10.3389/fenvc.2022.1060651

Vargas, C. A., Cuevas, L. A., Broitman, B. R., San Martin, V. A., Lagos, N. A., Gaitán-Espitia, J. D., & Dupont, S. (2022). Upper environmental pCO2 drives sensitivity to ocean acidification in marine invertebrates. *Nature Climate Change*, *12*(2), 200–207.

Vargas, C. A., Lagos, N. A., Lardies, M. A., Duarte, C., Manríquez, P. H., Aguilera, V. M., Broitman, B., Widdicombe, S., & Dupont, S. (2017). Species-specific responses to ocean acidification should account for local adaptation and adaptive plasticity. *Nature Ecology & Evolution*, *1*(4), 0084. https://doi.org/10.1038/s41559-017-0084

Wang, L., Wang, F., & Ma, Y. (2025). Combined effects of ocean acidification and copper exposure on the polyps of moon jellyfish Aurelia coerulea. *Ecotoxicology*, *34*(5), 712–724. https://doi.org/10.1007/s10646-025-02869-w

Wang, N., Wang, X. C., & Ma, X. (2015). Characteristics of concentration–inhibition curves of individual chemicals and applicability of the concentration addition model for mixture toxicity prediction. *Ecotoxicology and Environmental Safety*, *113*, 176–182. https://doi.org/10.1016/j.ecoenv.2014.12.008

Supplementary material

Table S1. Instrumental conditions for Li analysis by ICP-OES in seawater

Parameter	Value
Wavelength	670.79 nm
Plasma power	1300 W
Plasma gas	14 L/min
Auxiliary gas	0.5 L/min
Nebulizer gas	0.6 L/min
Radial offset	15 mm
Read time radial	1sec
Read time axial	3 sec
Peak evaluation	3 pixels

Table S2. Fits and statistics of the models fitted to growth rate data from single-driver experiments.

Driver	**Term**	**Estimate**	**Std. Error**	**t- value**	**p-value**	**Adj. R^2**	**Sig.**
Temp	(Intercept)	66.826	0.872	76.607	< 2e-16	0.7016	***
Temp	poly(temp, 2)1	30.827	6.586	4.681	1.96e-05		***
Temp	poly(temp, 2)2	-69.616	6.586	-10.57	9.26e-15		***
pH	(Intercept)	70.116	0.778	90.111	< 2e-16	0.3636	***
pH	poly(pH, 2)1	27.019	5.611	4.815	1.45e-05		***
pH	poly(pH, 2)2	-15.819	5.611	-2.819	0.00693		**
Lithium	(Intercept)	71.51	0.48	149	< 2e-16	0.2981	***
Lithium	poly(lithium, 2)1	-24.573	4.319	-5.689	2.14e-07		***
Lithium	poly(lithium, 2)2	-8.206	4.319	-1.9	0.0612		.

Table S3. Statistical tests comparing treatments in experiment #2 that have one driver out of reference conditions, and those that have more out of reference condition in addition to that one.

Condition	**Test**	**df**	**Statistic**	**p-value**
T targeted: 24 °C	Wilcoxon	-	W = 676	0.9722
pH targeted: 7.7	Welch's t-test	17.2	t = -3.211	0.0051
Li targeted: 7 ppm	Student's t-test	70	t = -1.197	0.2353

Table S4. Larvae counts

Sample	Larvae count
ST1	99.5
ST2	109
ST3	106.5
ST4	110.5
ST5	4.5
SP1	102.5
SP2	106
SP3	111.5
SP4	106.5
SP5	104.5
SL1	111
SL2	105
SL3	107
SL4	104.5
SL5	96.5
ML1P1_20_R1	98.5
ML1P1_20_R2	115.5
ML1P2_20_R1	110
ML1P2_20_R2	111.5
ML2P2_20_R1	107.5
ML2P2_20_R2	112.5
ML2P1_20_R1	115

ML2P1_20_R2	113.5
ML1P1_24_R1	101.5
ML1P1_24_R2	100
ML1P2_24_R1	129
ML1P2_24_R2	118
ML2P2_24_R1	99
ML2P2_24_R2	101
ML2P1_24_R1	123.5
ML2P1_24_R2	107

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